

Creating a Tetris Like Game:

Make all the basic Tetris shapes:

o, l, s, z, L, j, t

with differing colors

Make a start screen

- Title screen
 - o Quit button
 - o Play button

Make a game board screen

- Has the area where Tetris blocks are placed
- Part that tells the next block to the player
- Score indicator
- Play area is 10 width by 20 height

Game objective

To get as high as a score as possible

Score: based on how many lines have been cleared and Tetris blocks placed

Rules

- If a placed block goes above the game board, you lose
- If a line of blocks along the x-axis of the game board is placed, you will clear that line
- A block is 'placed' when one part of the bottom of the shape touches either the bottom or the game board or the top of another 'placed' block.
- When placed, it will not move, unless the blocks below it get cleared, then it will be translated downward by the number of blocks being cleared.
- Blocks being moved by the user (not 'placed') fall at a constant speed downward
- Blocks always start falling from the middle of the play area

Key features

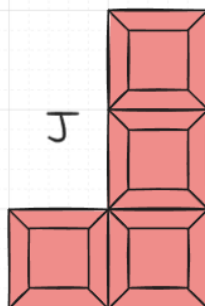
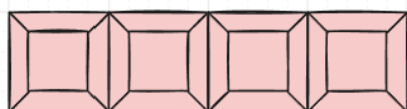
- Blocks will be able to be rotated
- Blocks can move left or right
- Blocks can speed downward
- Blocks can be dropped instantly downward
- Shapes will be randomly dropped based on the basic shapes (o, l, s, z, L, j, t)
- The speed of the blocks falling will increase based on the score achieved

Steps

- Implement a 10 x 20 grid array with each pixel representing a block
- Implement the Tetris blocks
- Track the current game state (start, playing, end)
- Implement movement, gravity, and time
- Implement collision detection
- Add inputs (arrow keys)
- Handle scoring logic
- Potentially make a nice start and end screen

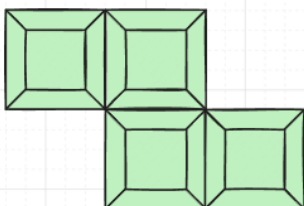
Illustrations:

- Each block will be a separate PNG that will be added to the JavaFX screen

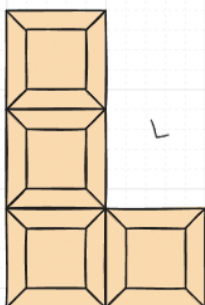


J

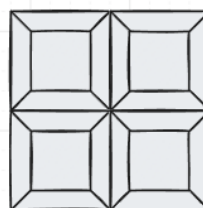
I



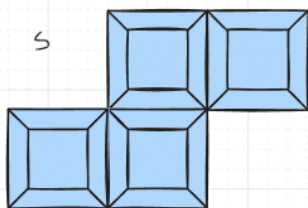
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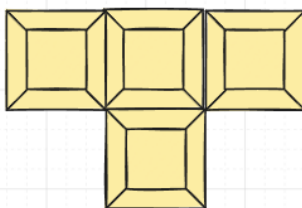
L



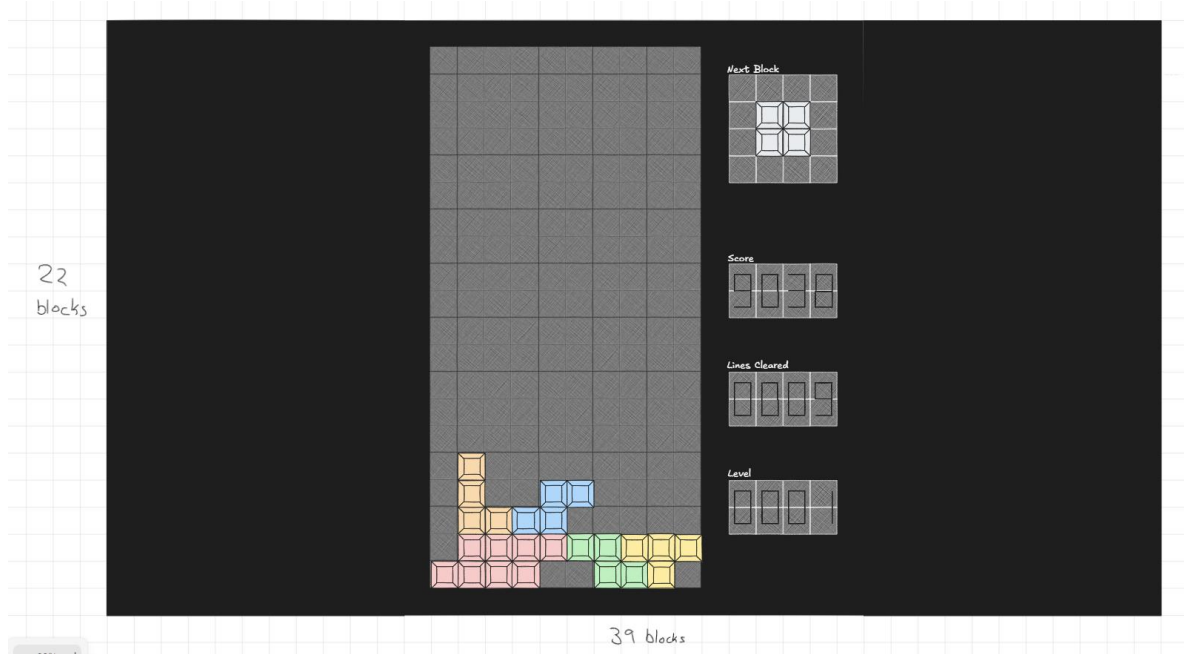
O



S



T



Should support 16:9 to work in full screen

Some very basic and hypothetical start and end screens:

